Semester Thesis – Introduction

Technical University Of Munich - Agile Robots SE

Carsten Oertel / 07.11.2025

Presentation Outline

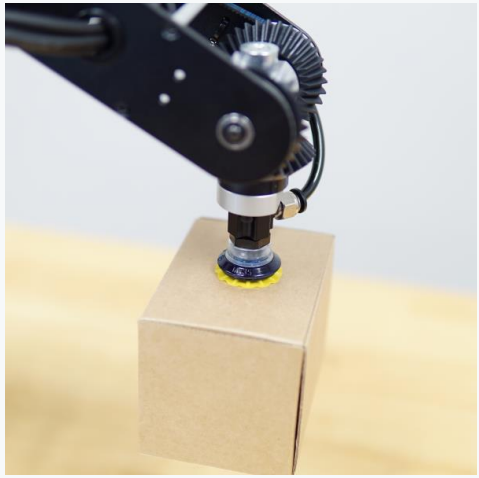
1. Motivation & Hypothesis
2. Problem Decomposition
3. State of the Art
4. Hardware & Sensing
5. Simulation & Learning Stack
6. Methods per Phase (Structure of research)
7. Evaluation Plan
8. Risks & Mitigations
9. Expected Contributions

01

Motivation and Hypothesis Statement

https://www.mybotshop.de/Franka-Emika-Franka-Research-3_1

https://dorna.ai/blog/types-of-grippers-for-robots/



https://dorna.ai/blog/types-of-grippers-for-robots/

Why:

- High-DoF hands are underused outside of labs
- Reliability in contact-rich tasks is the bottleneck

Question:

Can principled force-closure reasoning and finger-gaiting, possibly with a visuo-tactile policy, achieve robust and reliable thread start process of a screw using SoTA RL frameworks?

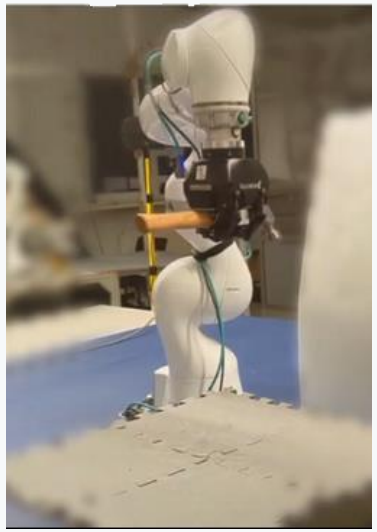
02

Problem Decomposition

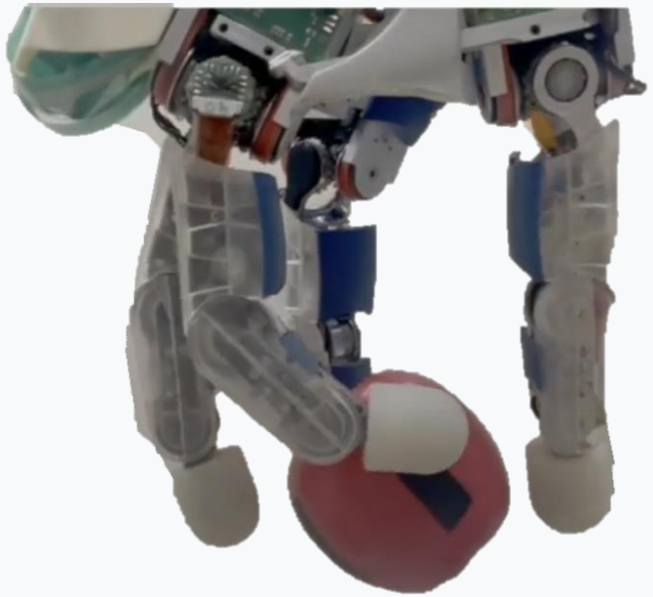
Problem Decomposition

grasping → finger-gaiting → micro alignment and thread start

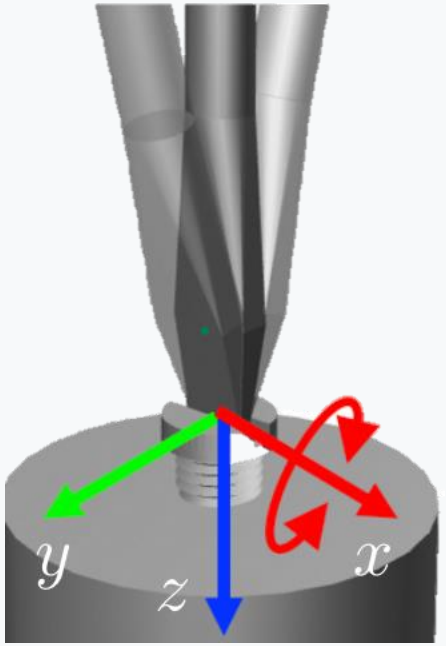
"Hammer" :
"Hand over"



https://david-s-martinez.github.io/DexGANGrasp.io/



https://aidx-lab.org/manipulation/humanoids23



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03

The Current State of the Art

State of the Art

- Dexterous grasping
- Dexterous manipulation
- Finger gaiting (force closure handling)
- Contact-rich tasks

Dexterous Grasping (task aware and generalizable)

UniDexGrasp / UniDexGrasp++ [1]:

- two-stage **proposal** → **goal-conditioned execution**; strong generalization across objects;
- ++ adds geometry-aware training and sizeable SR gains.

TaskDexGrasp [2]:

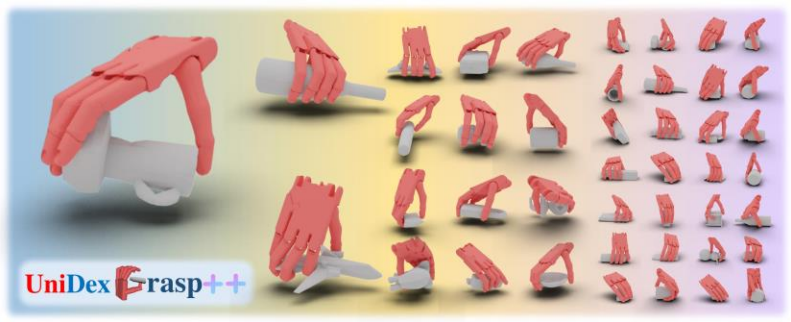
- **task-wrench–aware** synthesis (optimizes for downstream action forces/torques)
- better when grasps must enable later manipulation (e.g., thread-start).

DexGANGrasp [3]:

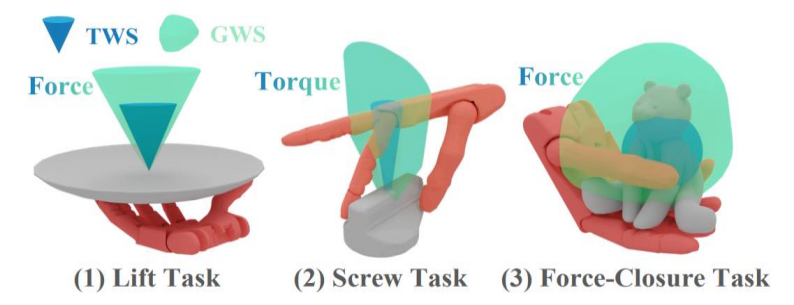
- fast single-view **generative** proposals with discriminator-style evaluator;
- attractive for real-time grasp candidate generation.

Our approach:

→ Use **UniDexGrasp++** as universal baseline; inject **task-wrench** constraints when preparing for micro-alignment & thread-start



https://github.com/PKU-EPIC/UniDexGrasp2



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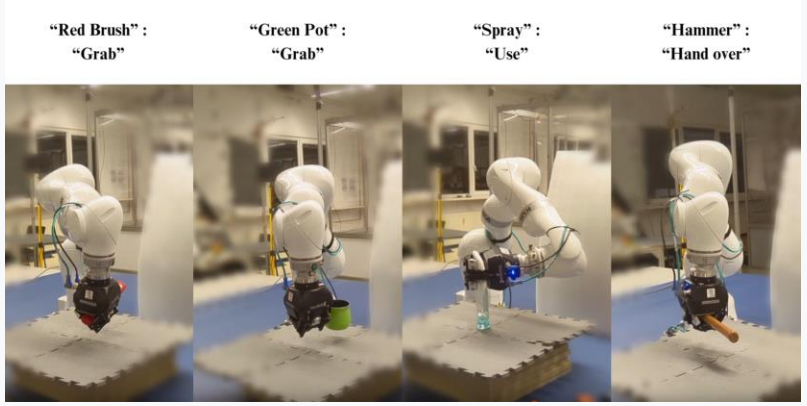
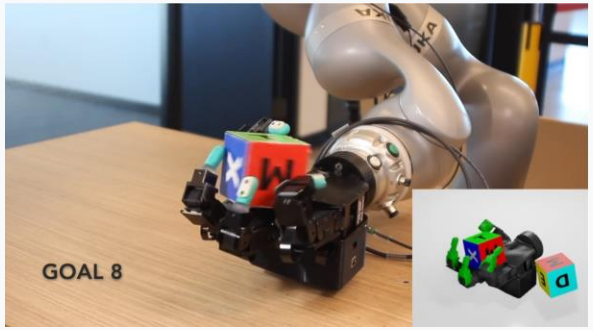


https://david-s-martinez.github.io/DexGANGrasp.io/

Dexterous Manipulation (no finger gaiting)

DeXtreme [4]:

- large-scale sim with domain randomization
- coupled policy + pose estimator
- robust **sim-to-real** on anthropomorphic hands



https://dextreme.org/

Sim-to-Real Vision-based Dexterous RL on Humanoids [5]:

- automated sim-to-real tuning → reducing sim-to-real gap
- generalized rewards
- clean **vision-only RL** recipe, no human demos.

https://arxiv.org/abs/2502.20396

DexPBT [6]:

- scalable Isaac-Gym training (population-based)
- includes reorientation and two-hand tasks.

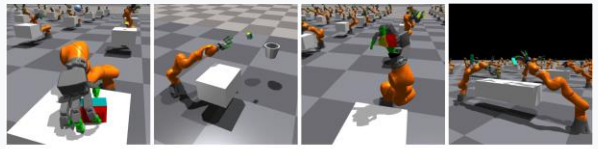


Fig. 1: Tasks trained with DexPBT. Left-to-right: *regrasping, throwing, single-handed reorientation, two-handed reorientation.*

https://arxiv.org/abs/2305.12127

Key insights

→ Strong transfer needs **privileged sim signals + robust estimator** at deploy time.

→ Works best if a **good initial grasp** is provided → reinforces pairing with a capable grasping stage.

Finger Gaiting or Force Closure Handling

Scoring with a RL-Critic [7]:

- use an **in-hand RL critic** to score/select grasps that maximize downstream in-hand performance → **bridges grasping ↔ manipulation**.

Time-Optimal & Speed-Adjustable Tactile In-Hand [8]:

- **tactile-only** reorientation; zero-shot to DLR Hand II
- preserves **force-closure**
- speed control without losing stability.

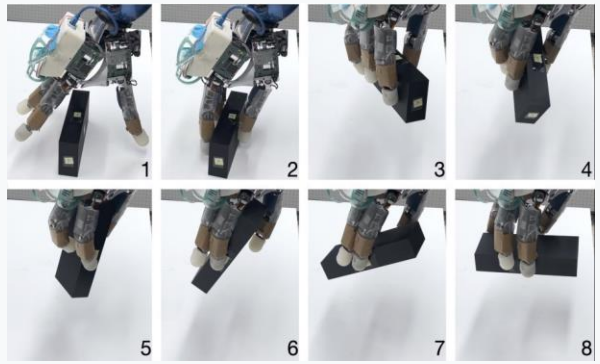
Shape-Conditioned Tactile Agent [9]:

- tactile policy conditioned on object shape class → better generalization with occluded vision.

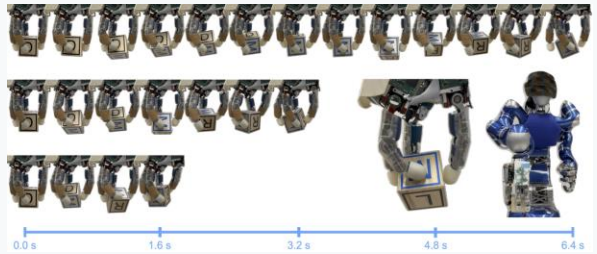
Key insights

→ **Goal-aware composition** outperforms isolated stages (pick the grasp your manipulator can exploit).

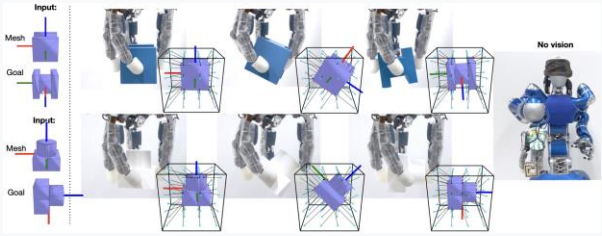
→ **Tactile-first** approaches deliver stability when vision is occluded and contact geometry matters (pre-thread-start).



https://arxiv.org/abs/2505.13253



https://arxiv.org/abs/2411.13148



https://arxiv.org/abs/2407.18834

Contact Rich tasks (Screwdriving and Screw Start)

Robotic Fastening with a Manual Screwdriver [10]:

- sequence **mount** → **capture** → **tighten**
  compliant spiral/contour search
- impedance to avoid slip-off
- clean failure detectors

Robotic Manipulation of Hand Tools: Screwdriving [11]:

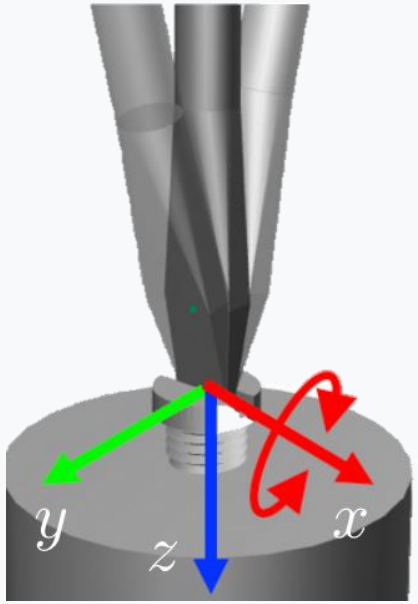
- **rolling contacts + finger gaits**
- pose estimation under contact
- **backward-chained force control**
- closest to dexterous hand + tool use

Physics-Informed Screwdriving [12]:

- system can accurately lean dynamics describing motion of screw tip
- robust to hole-pose uncertainty using
- incorporates physical thread/contact priors.

Key insights

→ Reliable thread-start hinges on coaxiality, axial preload, and recognizing the torque-angle & the axial-advance signatures



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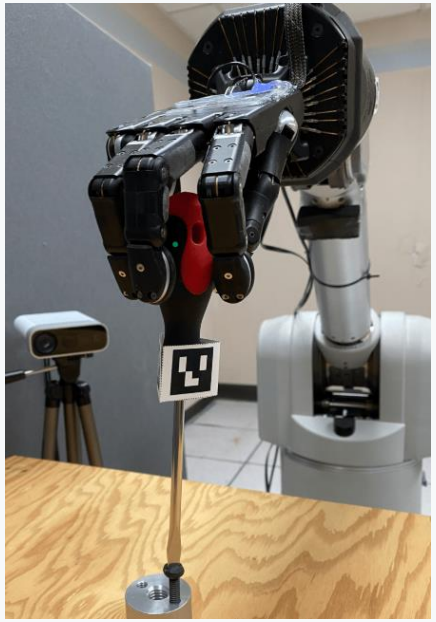

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Training Methods and Sim to Real Transfer

Model-Free (DeXtreme-style):

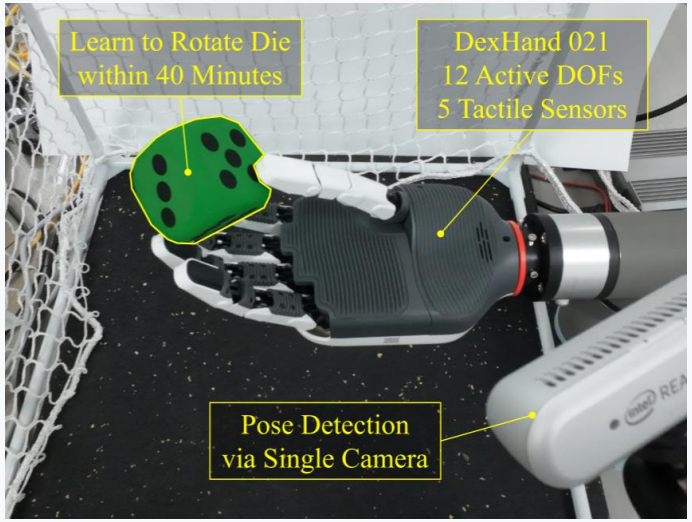
- excels with GPU-parallel sim + heavy DR; straightforward deployment once estimator is robust.
- Long training time

Probabilistic MBRL for Dexterous Hands [13]:

- ensembles + MPC reach multi-goal reorientation on a low-cost hand with **~80 min** real data.

Scaling toolkits:

- Isaac Lab/Gym, population-based training and domain randomization are now standard practice.



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04

Hardware and Sensors

Hardware & Sensors

Arms: 2× FR3; optional: Yu5 for heavier tasks.

Hands: Agile Hand(thumb abduction DoF).

Vision: RealSense eye-in-hand (D405 class for close range).

Tactile: fingertip sensors if needed (decision pending).

optionally:

05

Simulation and Learning Stack

Simulation and Learning Stack

ROS 2

Simulation

Training

Deployment

06

Methods and Planning

Methods and Planning

Phase 1

- Grasping and goal aware composition

  → perception unknown objects → UniDexGrasp ++ (grasp synthesis and execution) → critic base scorer

Phase 2

- Finger Gaiting

  → contact relocation → slip detection and recovery → arrival at pre-alingment pose with limited coaxial error

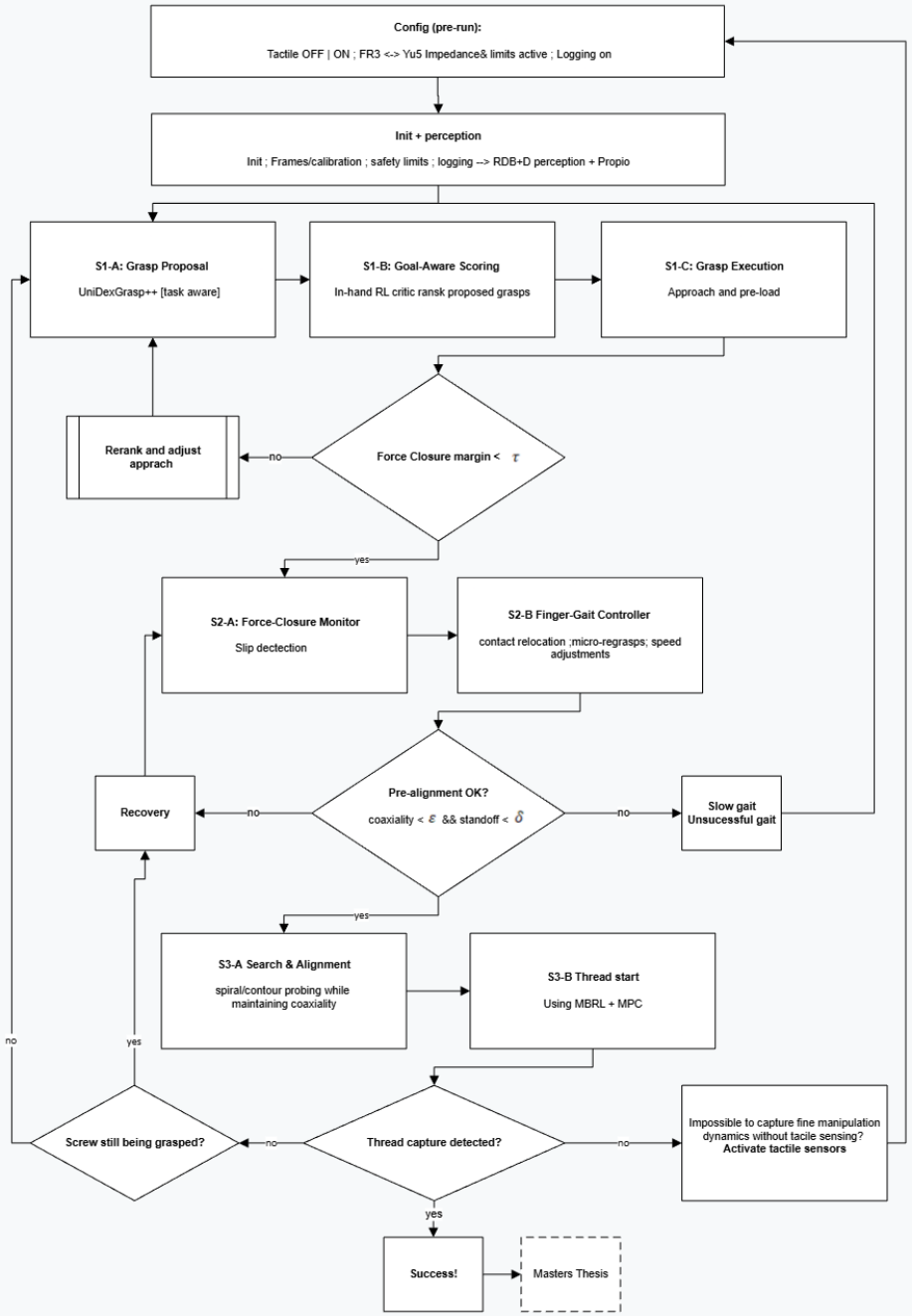
Phase 3

- Micro-Alignment & Thread Start

  → spiral and contour probing → fine alignment → gentle rotation

Sim to Real

Pipeline


Config (pre-run):
Tactile OFF | ON ; FR3 <-> Yu5 Impedance& limits active ; Logging on
Init + perception
Init ; Frames/calibration ; safety limits ; logging --> RDB+D perception + Propio
S1-A: Grasp Proposal
UniDexGrasp++ [task aware]
S1-B: Goal-Aware Scoring
In-hand RL critic ransk proposed grasps
S1-C: Grasp Execution
Approach and pre-load
Rerank and adjust apprach
no
Force Closure margin < τ
yes
S2-A: Force-Closure Monitor
Slip dectection
S2-B Finger-Gait Controller
contact relocation ;micro-regrasps; speed adjustments
Recovery
no
Pre-alignment OK?
coaxiality < ε && standoff < δ
no
Slow gait
Unsucessful gait
yes
S3-A Search & Alignment
spiral/contour probing while maintaining coaxiality
S3-B Thread start
Using MBRL + MPC
no
yes
Screw still being grasped?
no
Thread capture detected?
no
Impossible to capture fine manipulation dynamics without tacile sensing?
Activate tactile sensors
yes
Success!
Masters Thesis

07

Evaluation

Evaluation

Primary metric:

- Success rate of thread-start

Secondary:

- Time-to-align
- Peak contact forces/torques
- Number of slip events
- Pose error at engagement
- Repeatability
- Sim to real gap

Ablations:

- Grasping frameworks
- Vision and torque vs visuo-tactile

08

Risks and mitigations

Risks and mitigations

Hard thread modeling in sim

- emphasize real-robot micro-alignment early; treat sim as policy prior.

No wrist F/T

- rely on robot arm joint torques + impedance control
- perhaps adding tactile sensing if needed.

Hardware time

- Mechanical issues with agile hand mechanically sensitive to wrong assembly
- weekly validation slots

09

Contributions

Contributions

- Reusable grasp + gait stack with force-closure awareness,
- micro-alignment policy,
- Isaac/ROS2 assets,
- contact datasets

AGILE ROBOTS

Thank you

谢谢

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